

Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Technology, Pune 37
 (An Autonomous Institute affiliated to Savitribai Phule Pune University, Accredited with 'A++' Grade by NAAC)
Department of Engineering, Sciences and Humanities, A. Y. 2024-25
Name of the Course: Calculus and Statistics (HS)
Tutorial 11 on Unit 6-Random Variables

. Examples

1.	Compute the probability of obtaining at least two “six” in rolling a fair die 4 times.																
2.	If the chance that one of the ten telephone lines is busy at an instant is 0.2 a) What is the chance that 5 of the lines are busy? b) What is the most probable number of busy lines and what is the probability of this number? c) What is the probability that all the lines are busy?																
3.	If X is a random variable the difference between heads and tails obtained when a fair coin is tossed 3 times. What are the possible values of X and its probability mass function? Also write the distribution function of X .																
4.	A fair dice is rolled twice. Find the possible values of random variable X and its associated probability mass function, where X is the maximum of the values appearing in 2 rolls.																
5.	A random variable X takes values -3,-1, 2,5 with probabilities as shown in the table below. Determine the distribution of X .																
<table><tr><td>X</td><td>-3</td><td>-1</td><td>2</td><td>5</td></tr><tr><td>$P(X = x)$</td><td>$\frac{2k - 3}{10}$</td><td>$\frac{k - 1}{10}$</td><td>$\frac{k - 1}{10}$</td><td>$\frac{k - 2}{10}$</td></tr></table>		X	-3	-1	2	5	$P(X = x)$	$\frac{2k - 3}{10}$	$\frac{k - 1}{10}$	$\frac{k - 1}{10}$	$\frac{k - 2}{10}$						
X	-3	-1	2	5													
$P(X = x)$	$\frac{2k - 3}{10}$	$\frac{k - 1}{10}$	$\frac{k - 1}{10}$	$\frac{k - 2}{10}$													
6.	A random variable X has probability mass function (pmf) shown in the following tabular form. Find the value of unknown k . Hence write pmf and cdf of X . Draw graphs of pmf and cdf. Also find i) $p(1 \leq X < 3)$ ii) $p(1 < X \leq 3)$ iii) $p(X < 1)$ iv) $p(X > 5)$																
<table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$p(X = x)$</td><td>$k / 36$</td><td>$3k / 36$</td><td>$4k / 36$</td><td>$10k / 36$</td><td>$18k / 36$</td></tr></table>		X	1	2	3	4	5	$p(X = x)$	$k / 36$	$3k / 36$	$4k / 36$	$10k / 36$	$18k / 36$				
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$p(X = x)$	$k / 36$	$3k / 36$	$4k / 36$	$10k / 36$	$18k / 36$												
7.	A random variable X has the following p.m.f. <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$p(X = x_i)$</td><td>k</td><td>$3k$</td><td>$5k$</td><td>$7k$</td><td>$9k$</td><td>$11k$</td><td>$13k$</td></tr></table> Find (i) k (ii) $p(X \geq 2)$ (iii) $p(0 < X < 5)$ (iv) What is the minimum value of C for which $p(X \leq c) > 0.5$ (v) What is distribution function of X ?	X	0	1	2	3	4	5	6	$p(X = x_i)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$
X	0	1	2	3	4	5	6										
$p(X = x_i)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$										
8.	Determine k such that the following functions are p.m.f.s																

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	a) $P(x) = kx, \quad x = 1, 2, 3, \dots, 10$ b) $P(x) = k \frac{2^x}{x!}, \quad x = 0, 1, 2, 3$ c) $P(x) = k(2x^2 + 3x + 1), x = 0, 1, 2, 3$
9.	Verify whether the assignment $p(X = n) = 2^{-n}, n = 1, 2, 3, \dots$ is a probability mass function for random variable X .
10	A random variable X take four values with probabilities $\frac{1+3x}{4}, \frac{1-x}{4}, \frac{1+2x}{4}$ and $\frac{1-4x}{4}$. Find the condition on x that these values represent the probability mass function.
11	Let X be a continuous random variable with probability density function $F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{x}{2} & \text{if } 0 \leq x \leq 2 \\ 1 & \text{if } 2 < x \end{cases}$ Find: i) $p\left(\frac{1}{2} < X < \frac{3}{2}\right)$ (ii) $p(1 \leq X \leq 2)$
12	The time in years X required to complete a software project has a probability density function $f(x) = \begin{cases} kx(1-x), & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$ Compute the probability that the project will be completed in less than four months.
13	Let X be continuous random variable with probability density function $f(x) = \begin{cases} \frac{x}{2}, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$ Find (i) $p(1 \leq X \leq 1.5)$ (ii) $E(X)$ (iii) $\text{Var}(X)$ (iv) c.d.f.
14	Let X be a continuous random variable with probability density function $f(x) = \begin{cases} kx, & 0 \leq x \leq 2 \\ 2k, & 2 \leq x < 4 \\ -kx + 6k, & 4 \leq x < 6 \end{cases}$ Find k and the mean.
15	Verify $f(x) = 6x(1-x), 0 \leq x \leq 1$ is a probability density function for the diameter of a cable.

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	Find the condition on b such that $p(X < b) = p(X > b)$.
16	A continuous random variable X has pdf $f(x) = 3x^2, 0 \leq x < 1$. Find a such that $p(X > a) = 0.05$.
17	Let X be a continuous random variable with the following distribution. $F(x) = \begin{cases} kx, & \text{if } 0 \leq x \leq 5 \\ 0, & \text{elsewhere} \end{cases}$ (a) Find k. (b) Find $P(2 \leq X \leq 5)$. (c) Find c.d.f. (d) Plot the graph of p.m.f. and c.d.f.
18	Let X be a continuous random variable whose distribution f is constant on an interval, say $I = \{a \leq x \leq b\}$ and 0 elsewhere namely $f(x) = \begin{cases} k, & \text{if } a \leq x \leq b \\ 0, & \text{elsewhere} \end{cases}$ (a) Determine k. (b) Find the mean μ of X. (c) Determine the cumulative distribution function F of X.
19	Let X be a continuous random variable with distribution : $f(x) = \begin{cases} \frac{1}{6} + k, & \text{if } 0 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$ (a) Evaluate k. (b) Find $P(1 \leq X \leq 2)$.
20	Find the distribution function $f(x)$ of the continuous random variable X with the cumulative distribution function : (a) $F(x) = \begin{cases} 0, & \text{if } x < 0 \\ x^3, & \text{if } 0 \leq x \leq 1 \\ 1, & \text{if } x > 1 \end{cases}$ (b) $F(x) = \begin{cases} 0, & \text{if } x < 0 \\ \sin x, & \text{if } 0 \leq x \leq \frac{\pi}{2} \\ 1, & \text{if } x > \frac{\pi}{2} \end{cases}$
21	A box contains 8 items of which 2 are defective. A person draws 3 items from the box. Determine the expected number of defective items he has drawn.
22	A sample space of size 3 is selected at random from a box containing 12 items of which 3 are defective. If X denotes the number of defective items in the sample. Write the probability mass

	function and distribution function of X . Find the expected number of defective items.
23.	A player tosses two fair coins. The player wins \$2 if two heads occur, and \$1 if one head occur. On the other hand, the player losses \$3 if no heads occur. Find the expected gain of the player. Is the game fair?
24.	A fair coin is tossed until a head appears. What is the expectation of number of tosses.
25.	A box contains 8 items of which 2 are defective. A man draws 3 items from the box. Find the expected number of defective items he has drawn. (Ans: 3/4)
26.	Two dice are thrown simultaneously. Let X denote the sum of digits at the top. Find mathematical expectation of X .
27.	Two cards are drawn one by one with replacement from ten cards numbered 1 to 10. Find expectation of the product of the numbers on the drawn cards.
28.	The probability density function of a continuous random variable X is $f(x) = \begin{cases} x^3 & 0 < x \leq 1 \\ (2-x)^3 & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$ Find the expected value of the random variable.
29.	The probability density function of a continuous random variable X is $f(x) = ke^{- x }, \quad -\infty < x < \infty$ Find the value of k and also the mean and variance for the random variable.
30.	Suppose that X is a random variable for which $E(X) = 10$ and $V(X) = 25$. Find the positive values of a and b such that $Y = aX - b$ has expectation 0 and variance 1.
31.	Suppose that X is a random variable with mean μ and standard deviation σ . Find the expectation and variance of $f(x) = (x - \mu)/\sigma$.
32.	If it rains, a rain coat dealer can earn Rs 500 per day. If it is a dry day, he can lose Rs 100 per day. What is his expectation, if the probability of rain is 0.4?
33.	A box contains 4 white and 6 black ball. A man draws 2 balls and is given Rs 140 for every white ball and Rs 70 for every black ball. What is his expectation? (Ans:196)
34.	A player tosses two unbiased coins. He wins Rs 500 if 2 heads appear, Rs 200 if one head appear and Rs 100 if no head appears. Find the expected value of amount won by him.
35.	A player tosses two fair coins. He wins Rs 100 if head appears, Rs. 200 if 2 heads appear. On the other hand he loses Rs 500 if no head appears. Determine the expected value E of the game and is the game favorable to the player? (Ans $E=-25$, Not favorable)

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36.	A random variable X assumes n values $1, 2, \dots, n$ with equal probabilities. If the ratio of $\text{var}(X)$ to $E(X)$ is equal to 4, find the value of n .												
37.	Let a random variable has mean 2 and standard deviation $\frac{1}{2}$. Find i) $E(2X - 1)$ ii) $\text{Var}(X + 2)$ iii) $sd\left(\frac{3X - 1}{-4}\right)$												
38.	Given the following probability distribution: <table><tr><td>$X = x$</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>$P(X = x)$</td><td>0.15</td><td>0.30</td><td>0</td><td>0.30</td><td>0.25</td></tr></table> Find (i) $E(X)$ (ii) $E(2X + 3)$ (iii) $E(4X - 5)$ (iv) $E(X^2)$ (v) $\text{Var}(X)$ (vi) $\text{Var}(3X + 4)$	$X = x$	-2	-1	0	1	2	$P(X = x)$	0.15	0.30	0	0.30	0.25
$X = x$	-2	-1	0	1	2								
$P(X = x)$	0.15	0.30	0	0.30	0.25								